Department of Mathematics, IIT Madras
Linear Algebra for Engineers (MA2031)
Assignment -VI

1. Define $T : \mathbb{R}^2 \to \mathbb{R}^3$ by $T(a,b) = (a-b, a, 2a-b)$. Let $\mathcal{B}$ be the standard basis of $\mathbb{R}^2$. Take the ordered bases $\mathcal{C} = \{(1,2),(2,3)\}$ and $\mathcal{D} = \{(1,1,0),(0,1,1),(2,2,3)\}$ for $\mathbb{R}^3$. Compute $[T]_{\mathcal{B},\mathcal{D}}$ and $[T]_{\mathcal{C},\mathcal{D}}$.
2. Define $T : \mathbb{R}^3 \to \mathbb{R}^3$ by $T(a,b,c) = (b+c, c+a, a+b)$. Determine $[T]_{\mathcal{E},\mathcal{B}}$ where
   (a) $\mathcal{B} = \{(1,0,0),(0,0,1),(0,1,0)\}$, $\mathcal{E} = \{(0,0,1),(1,0,0),(0,1,0)\}$.
   (b) $\mathcal{B} = \{(1,1,-1),(-1,1,1),(1,-1,1)\}$, $\mathcal{E} = \{(-1,1,1),(1,-1,1),(1,1,-1)\}$.
3. Define $T : \mathbb{P}_2[t] \to \mathbb{R}$ by $T(f) = f(2)$. Compute the matrix of T with respect to the standard bases of the spaces.
4. Let $T : \mathbb{P}_2[t] \to \mathbb{P}_3[t]$ be given by $T(p(t)) = t\,p(t)$. Consider the ordered bases $\mathcal{B} = \{1+t,\ 1-t,\ t^2\}$ and $\mathcal{E} = \{1,\ 1+t,\ 1+t+t^2,\ t^3\}$ for $\mathbb{P}_2[t]$ and $\mathbb{P}_3[t]$, respectively. Find the matrix $[T]_{\mathcal{B},\mathcal{E}}$.
5. Let $B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$.
   (a) Define $T : \mathbb{R}^{2\times 2} \to \mathbb{R}^{2\times 2}$ by $T(A) = A^{\mathrm{t}}$. Compute $[T]_{\mathcal{B},\mathcal{B}}$.
   (b) Define $T : \mathbb{P}_2[t] \to \mathbb{R}^{2\times 2}$ by $T(f) = \begin{bmatrix} f'(0) & 2f(1) \\ 0 & f'(3) \end{bmatrix}$. Compute $[T]_{\mathcal{D},\mathcal{B}}$, where $\mathcal{D} = \{1,\ t,\ t^2\}$.
   (c) Define $T : \mathbb{R}^{2\times 2} \to \mathbb{R}$ by $T(A)$ = the sum of diagonal entries of A. Compute $[T]_{\mathcal{B},\{1\}}$.
6. Show that the map $T : \mathbb{F}^{n\times n} \to \mathbb{F}$ defined by $T(A) = \mathrm{tr}(A)$ is a linear functional. Show that T is an onto map. Also, find $nullity(T)$.
7. Consider the ordered bases $\mathcal{B}_1 = \{[1,0,1]^{\mathrm{t}},\ [1,1,0]^{\mathrm{t}},\ [0,1,1]^{\mathrm{t}}\}$ and $\mathcal{B}_2 = \{[1,-1,1]^{\mathrm{t}},\ [1,1,-1]^{\mathrm{t}},\ [-1,1,1]^{\mathrm{t}}\}$ for $\mathbb{R}^3$. Let T be the linear operator on $\mathbb{R}^3$ whose matrix representation with respect to the standard basis is given by the matrix $\begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Find the matrix $[T]_{\mathcal{B}_1,\mathcal{B}_2}$ and verify that $\left[T \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right]_{\mathcal{B}_2} = [T]_{\mathcal{B}_1,\mathcal{B}_2} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{\mathcal{B}_1}$.
8. Consider the ordered bases $\mathcal{B}_1 = \{u_1, u_2\}$ and $\mathcal{B}_2 = \{v_1, v_2\}$ for $\mathbb{F}^2$, where $u_1 = (1,1)^{\mathrm{t}}$, $u_2 = (-1,1)^{\mathrm{t}}$, $v_1 = (2,1)^{\mathrm{t}}$, and $v_2 = (1,0)^{\mathrm{t}}$. Let $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$.
   (a) Compute $Q = [A]_{\mathcal{B}_1,\mathcal{B}_1}$ and $R = [A]_{\mathcal{B}_2,\mathcal{B}_2}$.
   (b) Find the change of basis matrix $P = [I]_{\mathcal{B}_1,\mathcal{B}_2}$.
   (c) Verify the similarity relationship between Q and R.
9. Construct a matrix $A \in \mathbb{R}^{2\times 2}$, a vector $v \in \mathbb{R}^2$, and a basis $\mathcal{B} = \{u_1, u_2\}$ for $\mathbb{R}^2$ satisfying $[Av]_{\mathcal{B}} \neq A[v]_{\mathcal{B}}$.
10. Let $\{u_1, \ldots, u_n\}$ be an ordered basis of an inner product space V. Show the following:
    (a) The matrix $[a_{ij}]$, where $a_{ij} = \langle u_i, u_j \rangle$, is invertible. [This matrix is called the *Gram matrix* with respect to the given ordered basis.]

(b) If $(\alpha_1, \dots \alpha_n) \in \mathbb{F}^n$, then there is exactly one vector $x \in V$ such that $\langle x, u_j \rangle = \alpha_j$, for $j = 1, \dots, n$.

11. (a) Show that the eigenvalues of a triangular matrix (upper or lower) are the entries on the diagonal.
    (b) Show that for a real matrix A, the eigenvalues of A and A^T coincide.
    (c) Show that the sum of all the eigenvalues of A equals the sum of the diagonal entries of A and that the determinant of A equals the product of the eigenvalues.
12. Let A be an $n \times n$ matrix and let α be a scalar such that each row (or each column) sums to α. Show that α is an eigenvalue of A.
13. Let $T : V \to V$ be a linear operator. Prove the following:
    (a) If λ is an eigenvalue of T then λ^k is an eigenvalue of T^k.
    (b) If λ is an eigenvalue of T and $\alpha \in \mathbb{F}$, then $\lambda + \alpha$ is an eigenvalue of $T + \alpha I$.
    (c) Let $p(t) = a_0 + a_1 t + \dots + a_k t^k \in \mathbb{F}[t]$. If λ is an eigenvalue of T then $p(\lambda)$ is an eigenvalue of $p(T) := a_0 I + a_1 T + \dots + a_k T^k$.

    Are all the eigenvalues of $p(T)$ of the form $p(\lambda)$, where λ is an eigenvalue of T?
14. Let $T : V \to V$ be an isomorphism, where V is a finite dimensional vector space. Let λ be a nonzero scalar. Show that λ is an eigenvalue of T iff $1/\lambda$ is an eigenvalue of T^{-1}.
15. Let S and T be linear operators on V. Let λ and μ be eigenvalues of S and T, respectively. What is wrong with the following argument?

    $\lambda\mu$ an eigenvalue of $S \circ T$ because, if $S(x) = \lambda x$ and $T(x) = \mu x$, then $(S \circ T)x = S(\mu x) = \mu S(x) = \mu\lambda x = \lambda\mu x$.
16. Construct $A, B \in \mathbb{R}^{2\times 2}$ such that λ is an eigenvalue of A, μ is an eigenvalue of B but $\lambda\mu$ is not an eigenvalue of AB.
17. Find the characteristic polynomial, the eigenvalues and the associated eigenvectors for the matrices given below.

    (a) $\begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} -2 & -1 \\ 5 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} -2 & 0 & 3 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$
18. Find the eigenvalues and associated eigenvectors of the differentiation operator $d/dt : \mathbb{P}_3[t] \to \mathbb{P}_3[t]$.
19. Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a & b & c \end{bmatrix}$. Using Cayley-Hamilton theorem,
    (a) Show that A is invertible iff $a \neq 0$.
    (b) Show that if $a \neq 0$, then $A^{-1} = (1/a)(A^2 - cA - bI)$.
    (c) Express A^5 in terms of A^2, A, I.
20. Let A be a real symmetric matrix. Show that:
    (a) The eigenvalues of A are real.
    (b) Show that eigenvectors corresponding to distinct eigenvalues of A are orthogonal.
    (c) Let A be of order 3×3. Show that, if x and y are eigen vectors corresponding to distinct eigenvalues of A, then their cross product is a third eigenvector linearly independent with x and y.
21. Show that the eigenvalues of a unitary matrix lie on the unit circle.

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